Use approximate imperial conversions

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: __________________

1. Read the model: Use 5 miles is about 8 km. Estimate 30 miles in kilometres. Copy or complete the first step.

2. A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?

3. Convert 4.375 km to metres.

4. Miro says: Miro multiplies whenever converting, regardless of the direction between units. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Use 5 miles is about 8 km. Estimate 30 miles in kilometres. Copy or complete the first step.
   Answer About 48 km.

2. A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?
   Answer 31 complete pieces, with 10 cm left.

3. Convert 4.375 km to metres.
   Answer 4,375 m.

4. Miro says: Miro multiplies whenever converting, regardless of the direction between units. Is this correct? Explain.
   Answer Use the size of the target unit to decide the direction, then check whether the numerical value should grow or shrink.

Use approximate imperial conversions

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: __________________

1. Use 5 miles is about 8 km. Estimate 30 miles in kilometres.

2. A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?

3. Convert 4.375 km to metres.

4. Solve this independently and show every stage: Use 5 miles is about 8 km. Estimate 30 miles in kilometres.

5. Use a second method or representation for: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?

6. Check this result using an inverse, estimate or known fact: Convert 4.375 km to metres.

7. Prove or explain your reasoning: Convert 2.65 kg to grams and 3,750 ml to litres.

8. Identify and correct this misconception: Miro multiplies whenever converting, regardless of the direction between units.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Convert 4.375 km to metres.

Teacher answers and checking prompts

1. Use 5 miles is about 8 km. Estimate 30 miles in kilometres.
   Answer About 48 km.

2. A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?
   Answer 31 complete pieces, with 10 cm left.

3. Convert 4.375 km to metres.
   Answer 4,375 m.

4. Solve this independently and show every stage: Use 5 miles is about 8 km. Estimate 30 miles in kilometres.
   Answer About 48 km.

5. Use a second method or representation for: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?
   Answer Expected result: 31 complete pieces, with 10 cm left. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Convert 4.375 km to metres.
   Answer Expected result: 4,375 m. A valid check must be shown.

7. Prove or explain your reasoning: Convert 2.65 kg to grams and 3,750 ml to litres.
   Answer 2,650 g and 3.75 litres. The explanation must show why the method works.

8. Identify and correct this misconception: Miro multiplies whenever converting, regardless of the direction between units.
   Answer Use the size of the target unit to decide the direction, then check whether the numerical value should grow or shrink.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Convert 4.375 km to metres.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Use approximate imperial conversions

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. Prove or explain your reasoning: Convert 2.65 kg to grams and 3,750 ml to litres.

2. Prove the answer to this question, rather than only calculating it: Use 5 miles is about 8 km. Estimate 30 miles in kilometres.

3. Solve this in two different ways and compare them: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?

4. Work backwards from the answer to reconstruct the question: 4,375 m.

5. Explain why this reasoning fails: Miro multiplies whenever converting, regardless of the direction between units.

6. Create a new example that tests the same idea as: Convert 4.375 km to metres.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Convert 2.65 kg to grams and 3,750 ml to litres.
   Answer 2,650 g and 3.75 litres. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: Use 5 miles is about 8 km. Estimate 30 miles in kilometres.
   Answer Expected result: About 48 km. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?
   Answer Expected result: 31 complete pieces, with 10 cm left. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 4,375 m.
   Answer One valid reconstruction is: Convert 4.375 km to metres. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro multiplies whenever converting, regardless of the direction between units.
   Answer Use the size of the target unit to decide the direction, then check whether the numerical value should grow or shrink.

6. Create a new example that tests the same idea as: Convert 4.375 km to metres.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.